
GEOODE-KD: DISTILLING SENTENCE EMBEDDINGS AS TEACHER-GUIDED GEOMETRIC DYNAMICS

Anonymous authors

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ABSTRACT

Embedding distillation can become restrictive when intermediate supervision treats representations at different depths as independent matches to the same final teacher embedding. Such objectives specify target states, but leave the transitions between them undefined. We propose **GeoODE-KD**, a sentence embedding distillation framework that turns cached teacher outputs into depth-dependent transition targets. GeoODE-KD places normalized embeddings on a hypersphere and defines a teacher-conditioned potential from point-wise semantic alignment and native teacher geometry. Its negative Riemannian gradient gives the direction of semantic change, while a finite-horizon schedule determines how much of the remaining displacement each student layer should cover. GeoODE-KD does not insert a Neural ODE module or run a numerical solver; instead, it treats the existing Transformer layers as a discrete trajectory and trains every intermediate transition to move along the teacher-conditioned vector field. The method uses only final teacher embeddings cached before training and leaves the inference architecture unchanged. Its ideal continuous flow preserves embedding norms and decreases the teacher-conditioned energy; without the relational term, geodesic distance to the teacher contracts in closed form and reaches zero at the final depth. Across three teacher-student pairs and nine benchmarks, students distilled on 102,361 unlabeled sentences recover 97–98% of their teacher’s average score, and improve on the strongest multi-layer anchoring baseline by 3.0 and 2.2 points on the two MiniLM students while trailing it by 0.8 points on BERT-base.

1 INTRODUCTION

Dense text embeddings support semantic search, retrieval-augmented generation, clustering, classification, and semantic textual similarity. The strongest embedding models, however, are often too large or expensive for routine deployment. Knowledge distillation addresses this cost by training a compact student from a stronger teacher (Hinton et al., 2015; Reimers & Gurevych, 2020). Most embedding distillation pipelines still treat this transfer as a state-matching problem: given a teacher embedding, make one or more student representations resemble it.

Prior work improves state matching in several ways, but each leaves the path through student depth unspecified. Output-level methods align the final embedding and add contrastive supervision (Xu et al., 2023; Zhang et al., 2024). Relational and optimal-transport objectives also transfer pairwise or token-level structure (Truong et al., 2025). These methods provide richer terminal supervision, but they do not determine how intermediate sentence representations should change. Depth-aware methods supervise several student layers or propagate structure between adjacent layers (Li et al., 2025; Dao et al., 2026). They expose more of the student to the teacher signal, but selected layers are still trained as separate target states.

This formulation becomes restrictive when teacher and student have different capacities. A shallow student layer must retain information needed by later computation, while a final teacher embedding already represents the end of a much deeper transformation. Pulling many shallow layers toward that same endpoint can flatten the student’s internal hierarchy: the model is asked to produce terminal semantics before it has completed the computation needed to support them. Choosing fewer anchor layers reduces the constraint, but still does not specify what should happen between two anchors.

The missing signal is not another target state; it is a rule for moving from the current state to the next one.

We propose **GeoODE-KD**, a sentence embedding distillation framework that makes this transition rule explicit. GeoODE-KD views normalized layer representations as points on a hypersphere and uses cached teacher embeddings to define a semantic potential. A point-wise term attracts each sentence to its teacher target, while a relational term preserves the teacher’s native cosine geometry. The negative Riemannian gradient of this potential gives a local direction of semantic improvement. A finite-horizon schedule then assigns each student layer a fraction of the displacement that remains before the output.

GeoODE-KD does not replace Transformer blocks with a Neural ODE or train a separate velocity network. Instead, it reads the existing Transformer layers as a discrete trajectory. At depth t_ℓ , the teacher-conditioned field gives the direction in which the state should move toward $t_{\ell+1}$. The next Transformer layer is trained so that its realized update points along this field. Intermediate layers are therefore supervised by transitions, $\mathbf{Z}^{(\ell)} \rightarrow \mathbf{Z}^{(\ell+1)}$, rather than independent copies of the terminal condition, $\mathbf{Z}^{(\ell)} \rightarrow \mathbf{T}$. This use of continuous dynamics is a training principle, not a change to the student architecture (Chen et al., 2018; Li et al., 2022).

The construction also separates teacher cost from student optimization. The teacher is run once, and only its final sentence embeddings are cached. GeoODE-KD requires no teacher hidden states, attention maps, learned projection head, or numerical ODE solve during training. The geometric machinery is discarded after training, leaving the original student encoder at inference. The formulation further makes the claimed behavior measurable across depth: the ideal flow preserves embedding norms and decreases teacher-conditioned energy, while its instance-only form follows a closed geodesic contraction profile that reaches the teacher at the final depth.

Our contributions are threefold:

- We formulate embedding distillation as trajectory supervision, where the teacher specifies transitions between consecutive student layers rather than a collection of independent target states.
- We derive a finite-horizon Riemannian field that combines point-wise alignment with native teacher geometry and produces an explicit local target at every student depth.
- We train the existing Transformer layers to follow these targets with a velocity-matching objective that uses only cached teacher outputs and adds no inference-time parameters.

2 BACKGROUND AND RELATED WORK

2.1 STATIC EMBEDDING DISTILLATION

Let $\mathbf{z}_i^{(\ell)}$ be the normalized representation of sentence x_i after student layer ℓ , and let τ_i be its normalized teacher target. Endpoint distillation constrains only

$$\mathbf{z}_i^{(L)} \longrightarrow \tau_i. \quad (1)$$

This objective is efficient, but leaves the internal path unconstrained. Multi-layer distillation adds the same target at selected depths $\mathcal{A}$,

$$\mathcal{L}_{\text{layer}} = \sum_{\ell \in \mathcal{A}} D(\mathbf{Z}^{(\ell)}, \mathbf{T}), \quad (2)$$

where $\mathbf{Z}^{(\ell)}$ and $\mathbf{T}$ stack a mini-batch of student and teacher embeddings. Relational variants add discrepancies between pairwise similarity matrices. All three cases supervise states: endpoint KD constrains one state, multi-layer KD constrains several states, and relational KD constrains their geometry.

Equation 2 does not say how much semantic change is appropriate at layer ℓ . Weak supervision may have little effect before the output; strong supervision may force shallow features to imitate a state that the teacher produces only after substantially more computation. Selecting fewer anchors reduces this tension but still leaves the transitions undefined.

2.2 DEPTH AS SEMANTIC TIME

We index student representations by normalized depth,

$$t_\ell = \frac{\ell}{L}, \quad \mathbf{Z}^{(\ell)} \approx \mathbf{Z}(t_\ell), \quad (3)$$

and treat consecutive layers as observations along a semantic trajectory. A teacher-conditioned vector field then specifies the local transition,

$$\frac{d\mathbf{Z}(t)}{dt} = F(\mathbf{Z}(t), \mathbf{T}, t). \quad (4)$$

Static matching asks whether $\mathbf{Z}^{(\ell)}$ resembles the terminal teacher state. Equation 4 instead asks whether the transition $\mathbf{Z}^{(\ell)} \rightarrow \mathbf{Z}^{(\ell+1)}$ moves in a direction and at a pace that reduce teacher discrepancy.

Semantic time is an ordered coordinate for supervision; it is not an assumption that a pretrained Transformer is literally the solution of a latent ODE. The student remains a conventional finite-depth encoder. Continuous dynamics only provide compatible local targets whose composition leads toward the endpoint.

2.3 RELATED DISTILLATION PARADIGMS

Sentence-embedding distillation. Direct regression is attractive because teacher sentence embeddings can be cached and reused (Reimers & Gurevych, 2020). Contrastive and multi-stage methods enrich this signal (Xu et al., 2023; Zhang et al., 2024); hidden-state relations and optimal transport address architectural or tokenizer mismatch (Truong et al., 2025). These methods strengthen what is matched. GeoODE-KD instead changes how the target is used: each cached output defines both an endpoint and a direction of evolution.

Layer-wise and geometric transfer. ESE trains depth-scalable representations, while TALAS combines upper-layer teacher anchors with adjacent-layer structural propagation (Li et al., 2025; Dao et al., 2026). Both make depth explicit, but supervise states or relations at selected locations. GeoODE-KD derives each local transition from one teacher-conditioned potential, so relational knowledge changes the direction of motion rather than acting only as another state constraint.

Continuous-depth and flow-based transfer. Neural ODEs replace discrete blocks with a learned differential equation, while ODE-inspired Transformers use integration rules in architecture design (Chen et al., 2018; Li et al., 2022). Diffusion and flow matching also transport student features or distributions toward a teacher (Huang et al., 2023; Lipman et al., 2023; Chen & Lipman, 2024; Shao et al., 2024). GeoODE-KD learns no velocity network and performs no continuous solve. Its field is the analytic gradient of a teacher-conditioned potential, and the existing Transformer layers follow a geometric discretization of that field. The continuous formulation therefore disappears at inference time.

3 GEOODE-KD: FROM TEACHER EMBEDDINGS TO SEMANTIC DYNAMICS

Given an unlabeled corpus $\mathcal{D} = \{x_i\}_{i=1}^N$, a teacher f_T , and a student f_S with L Transformer layers, our goal is to train the student without labels or online teacher inference. GeoODE-KD follows four steps. It first caches and maps teacher embeddings into the student space. It then converts teacher discrepancy into a tangent vector field, assigns this field a finite depth horizon, and uses the resulting local updates to supervise consecutive student layers. All four steps define training targets; none adds a module to the student.

3.1 TEACHER TARGET CONSTRUCTION

Let

$$f_T : \mathcal{X} \rightarrow \mathbb{R}^{d_T}, \quad f_S : \mathcal{X} \rightarrow \mathbb{R}^{d_S} \quad (5)$$

denote the teacher and student embedding functions. The offline stage runs f_T once and caches

$$\tilde{\tau}_i = \text{norm}(f_T(x_i)), \quad \text{norm}(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|_2 + \epsilon}. \quad (6)$$

No teacher hidden state or attention map is retained.

Point-wise supervision requires both models to use the same width. For $d_T \geq d_S$, we fit a fixed PCA basis $P_{\text{PCA}} \in \mathbb{R}^{d_T \times d_S}$ on the cached training embeddings. This basis selects a maximum-variance d_S -dimensional teacher subspace, but its orientation is arbitrary. The orientation leaves cosine geometry unchanged, yet a point-wise loss would still force the pretrained student to match it.

We fix this coordinate *gauge* before training. Let $\mathbf{T}_{\text{PCA}}$ contain normalized PCA coordinates for a fitting subset, and let $\mathbf{Z}_0^{(L)}$ contain normalized output embeddings of the initial student for the same sentences. We solve orthogonal Procrustes (Schönemann, 1966):

$$R^* = \arg \min_{R \in O(d_S)} \left\| \mathbf{T}_{\text{PCA}} R - \mathbf{Z}_0^{(L)} \right\|_F, \quad R^* = UV^\top, \quad U\Sigma V^\top = \text{svd}\left(\mathbf{T}_{\text{PCA}}^\top \mathbf{Z}_0^{(L)}\right). \quad (7)$$

The shared-space point-wise target is then

$$\tau_i = \text{norm}(f_T(x_i) P_{\text{PCA}} R^*) \in \mathbb{S}^{d_S-1}. \quad (8)$$

When $d_T = d_S$, P_{PCA} can be the identity. The rotation R^* changes neither pairwise cosines nor retained variance. We freeze both maps after fitting; Appendix B gives the target construction details.

For the student, let $\mathbf{h}_i^{(\ell)}$ be the token states after Transformer layer ℓ . Pooling and normalization produce

$$\mathbf{z}_i^{(\ell)} = \text{norm}\left(\text{Pool}(\mathbf{h}_i^{(\ell)})\right) \in \mathbb{S}^{d_S-1}. \quad (9)$$

For a mini-batch of B sentences, we stack these row-wise into $\mathbf{Z}^{(\ell)}, \mathbf{T} \in \mathbb{R}^{B \times d_S}$.

Point-wise alignment uses the projected targets $\mathbf{T}$, but pairwise cosines do not require equal widths. For each mini-batch, we construct the native teacher Gram matrix

$$\mathbf{G}_T = \tilde{\mathbf{T}} \tilde{\mathbf{T}}^\top, \quad (10)$$

where $\tilde{\mathbf{T}}$ stacks the embeddings in Eq. 6. This matrix is independent of teacher width and invariant to R^* , so projection affects only the point-wise target.

3.2 TEACHER-CONDITIONED VECTOR FIELD

The next stage turns teacher discrepancy into a direction of motion. For a student state $\mathbf{z}_i$ and projected target τ_i , the hyperspherical distance is

$$d_g(\mathbf{z}_i, \tau_i) = \arccos(\mathbf{z}_i^\top \tau_i). \quad (11)$$

The point-wise potential averages its squared value,

$$\mathcal{E}_{\text{sem}}(\mathbf{Z}, \mathbf{T}) = \frac{1}{2B} \sum_{i=1}^B d_g^2(\mathbf{z}_i, \tau_i). \quad (12)$$

Its negative Riemannian gradient points along the shortest geodesic to the teacher target.

The relational potential compares the student Gram matrix with the native teacher matrix from Eq. 10:

$$\mathcal{E}_{\text{geo}}(\mathbf{Z}, \mathbf{G}_T) = \frac{1}{B^2} \left\| \mathbf{Z} \mathbf{Z}^\top - \mathbf{G}_T \right\|_F^2. \quad (13)$$

We combine the two terms as

$$\mathcal{E}(\mathbf{Z}; \mathbf{T}, \mathbf{G}_T) = \alpha \mathcal{E}_{\text{sem}}(\mathbf{Z}, \mathbf{T}) + \beta \mathcal{E}_{\text{geo}}(\mathbf{Z}, \mathbf{G}_T). \quad (14)$$

with $\alpha, \beta \geq 0$. GeoODE-KD does not minimize this expression separately at every layer. It differentiates the potential to obtain the next transition target.

For $\mathbf{z} \in \mathbb{S}^{ds-1}$, the projection of an ambient vector $\mathbf{u}$ onto the tangent space is

$$\Pi_{\mathbf{z}}(\mathbf{u}) = \mathbf{u} - (\mathbf{z}^\top \mathbf{u})\mathbf{z}. \quad (15)$$

The hyperspherical logarithmic map is

$$\text{Log}_{\mathbf{z}}(\boldsymbol{\tau}) = \frac{\theta}{\sin \theta} \Pi_{\mathbf{z}}(\boldsymbol{\tau}), \quad \theta = d_g(\mathbf{z}, \boldsymbol{\tau}). \quad (16)$$

Applying both maps row-wise gives

$$\mathbf{V}(\mathbf{Z}, \mathbf{T}, \mathbf{G}_T) = \alpha \text{Log}_{\mathbf{Z}}(\mathbf{T}) - \frac{4\beta}{B} \Pi_{\mathbf{Z}}[(\mathbf{Z}\mathbf{Z}^\top - \mathbf{G}_T)\mathbf{Z}]. \quad (17)$$

This direction equals $-B \text{grad}_{\mathbb{S}} \mathcal{E}$. The factor B removes the scale introduced by batch averaging, so flow speed does not change with mini-batch size.

3.3 FINITE-HORIZON DEPTH SCHEDULE

A standard gradient flow approaches its target asymptotically, while the student has only L layers. We map these layers to $t \in [0, 1]$. Let $s(t) \geq 0$ allocate guidance over depth, with remaining mass

$$R(t) = \int_t^1 s(u) du. \quad (18)$$

The depth-indexed field is

$$\frac{d\mathbf{Z}(t)}{dt} = \frac{s(t)}{R(t)} \mathbf{V}(\mathbf{Z}(t), \mathbf{T}, \mathbf{G}_T), \quad (19)$$

for $t < 1$. The ratio $s(t)/R(t)$ maps unbounded gradient-flow time into this finite interval. We use $s(t) = t$, which gives $R(t) = (1 - t^2)/2$ and places more guidance near the output. Constant and power schedules are direct alternatives.

The instance-only case shows what this schedule controls. For $\alpha = 1$ and $\beta = 0$, let $\theta(t) = d_g(\mathbf{z}(t), \boldsymbol{\tau})$. The log map has norm $\theta(t)$ and points along the target geodesic, so

$$\dot{\theta}(t) = -\frac{s(t)}{R(t)} \theta(t), \quad \theta(t) = \theta(0) \frac{R(t)}{R(0)}. \quad (20)$$

With $s(t) = t$, this becomes $\theta(t) = \theta(0)(1 - t^2)$ and $\theta(1) = 0$. Exact arrival applies to the instance-only field. When $\beta > 0$, the relational term couples examples and may bend individual paths; the joint potential still decreases under the ideal continuous dynamics (Section 3.5).

3.4 LAYER-WISE FLOW SUPERVISION

During training, the L Transformer layers provide states at $t_\ell = \ell/L$; no external solver evaluates Eq. 19. The fraction of remaining point-wise displacement assigned to transition ℓ is

$$\begin{aligned} \rho_\ell &= 1 - \exp\left(-\int_{t_\ell}^{t_{\ell+1}} \frac{s(u)}{R(u)} du\right) \\ &= 1 - \frac{R(t_{\ell+1})}{R(t_\ell)} \in (0, 1]. \end{aligned} \quad (21)$$

This integrated form remains well-defined on the final interval and gives $\rho_{L-1} = 1$ for any valid schedule.

At layer ℓ , we evaluate the field once and construct the next target as

$$\tilde{\mathbf{Z}}^{(\ell+1)} = \text{Exp}_{\mathbf{Z}^{(\ell)}}\left(\rho_\ell \mathbf{V}(\mathbf{Z}^{(\ell)}, \mathbf{T}, \mathbf{G}_T)\right). \quad (22)$$

For tangent vector $\mathbf{v}$, the spherical exponential map is

$$\text{Exp}_{\mathbf{z}}(\mathbf{v}) = \cos(\|\mathbf{v}\|_2)\mathbf{z} + \sin(\|\mathbf{v}\|_2) \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \quad (23)$$

with its continuous extension at $\mathbf{v} = \mathbf{0}$. This update remains on the hypersphere for any local step size. For $\beta = 0$ and $\alpha = 1$, it is spherical interpolation toward the teacher by fraction ρ_ℓ . With the relational term, it is a one-step discretization of the coupled field, not an exact solution of the nonlinear trajectory.

The Transformer produces the actual state $\mathbf{Z}^{(\ell+1)}$. On the hypersphere, the actual movement made by the student from layer ℓ to $\ell + 1$ is the tangent vector

$$\mathbf{U}^{(\ell)} = \text{Log}_{\mathbf{Z}^{(\ell)}}(\mathbf{Z}^{(\ell+1)}), \quad (24)$$

and the direction toward the teacher endpoint from the same state is

$$\mathbf{V}^{(\ell)} = \text{Log}_{\mathbf{Z}^{(\ell)}}(\mathbf{T}), \quad \mathbf{v}_i^{(\ell)} = \text{Log}_{\mathbf{z}_i^{(\ell)}}(\boldsymbol{\tau}_i). \quad (25)$$

The endpoint-guided velocity loss asks each realized transition to point along this direction:

$$\mathcal{L}_{\text{evel}} = \frac{1}{L-1} \sum_{\ell=1}^{L-1} \frac{1}{B} \sum_{i=1}^B \left[1 - \cos(\mathbf{u}_i^{(\ell)}, \mathbf{v}_i^{(\ell)}) \right], \quad (26)$$

where $\mathbf{u}_i^{(\ell)}$ and $\mathbf{v}_i^{(\ell)}$ are the rows of $\mathbf{U}^{(\ell)}$ and $\mathbf{V}^{(\ell)}$ and $\cos(\cdot, \cdot)$ is the cosine between tangent vectors. Both vectors live in the tangent space of $\mathbf{Z}^{(\ell)}$, so the comparison is well defined. The cosine is scale-free, so $\mathcal{L}_{\text{evel}}$ encourages each student layer to move in the tangent direction toward the final teacher embedding without prescribing an intermediate target state or a step magnitude. Every transition is covered, including the last one $\ell = L - 1$ into the final layer: the endpoint objective below fixes *where* that layer must land, while $\mathcal{L}_{\text{evel}}$ asks that it be reached along the geodesic rather than by an arbitrary detour. In practice the direction is computed from a detached $\mathbf{Z}^{(\ell)}$, so the loss cannot be lowered by moving the source state to reorient the target.

3.5 TRAINING OBJECTIVE

The final layer must still satisfy the endpoint condition. We use

$$\mathcal{L}_{\text{end}} = \frac{1}{B} \sum_{i=1}^B \left(1 - (\mathbf{z}_i^{(L)})^\top \boldsymbol{\tau}_i \right). \quad (27)$$

No earlier layer receives this direct target. The direction $\mathbf{v}_i^{(L-1)}$ points along the geodesic to $\boldsymbol{\tau}_i$, so $\mathcal{L}_{\text{end}}$ is the boundary condition of the same path: $\mathcal{L}_{\text{evel}}$ orients the final transition and $\mathcal{L}_{\text{end}}$ fixes where it must stop.

We also apply unsupervised contrastive learning at the output. For two dropout views $\mathbf{z}_i^{(L,a)}$ and $\mathbf{z}_i^{(L,b)}$,

$$\mathcal{L}_{\text{ctr}} = -\frac{1}{B} \sum_{i=1}^B \log \frac{\exp((\mathbf{z}_i^{(L,a)})^\top \mathbf{z}_i^{(L,b)} / \tau_c)}{\sum_{j=1}^B \exp((\mathbf{z}_i^{(L,a)})^\top \mathbf{z}_j^{(L,b)} / \tau_c)}. \quad (28)$$

Restricting this term to the output avoids a second layer-wise objective. The training loss is

$$\mathcal{L}_{\text{total}} = \lambda_{\text{end}} \mathcal{L}_{\text{end}} + \lambda_{\text{evel}} \mathcal{L}_{\text{evel}} + \lambda_{\text{ctr}} \mathcal{L}_{\text{ctr}}. \quad (29)$$

Here, λ_{end} , λ_{evel} , and λ_{ctr} weight endpoint, transition, and contrastive supervision. The potential $\mathcal{E}$ and its flow describe the ideal trajectory and drive the depth diagnostics; neither is an additional layer-wise loss. The term $\mathcal{L}_{\text{evel}}$ measures whether each Transformer layer moves along the geodesic toward the teacher endpoint.

Properties of the ideal flow. The construction gives three properties. First, the field is tangent to the product of hyperspheres. Consequently, $d\|\mathbf{z}(t)\|_2^2/dt = 2\mathbf{z}(t)^\top \dot{\mathbf{z}}(t) = 0$, so continuous trajectories preserve embedding norms. Second, for $t < 1$ the teacher-conditioned potential decreases monotonically:

$$\frac{d\mathcal{E}}{dt} = -\frac{B s(t)}{R(t)} \|\text{grad}_{\mathcal{S}} \mathcal{E}\|_F^2 \leq 0. \quad (30)$$

Third, Eq. 20 gives exact geodesic contraction for the instance-only field. These are properties of the ideal continuous flow; whether finite Transformer layers realize them is measured empirically. Full derivations are given in Appendix A.

Training and inference. Across L layers, the Gram term costs $\mathcal{O}(LB^2d_S)$ training operations and $\mathcal{O}(B^2)$ temporary memory, independent of teacher size. Target construction and geometric supervision are removed at deployment, so inference uses the original student architecture and parameter count. Implementation details are provided in Appendix C.

4 EXPERIMENTS

4.1 SETUP

Teacher-student pairs. We evaluate three teacher-student configurations that differ in teacher family, teacher width, and student capacity: (a) Qwen3-Embedding-4B ($d_T = 2560$, last-token pooling) $\rightarrow$ BERT-base (109M, $d_S = 768$, 12 layers), the capacity-gap setting; (b) BGE-M3 ($d_T = 1024$, an XLM-R encoder with CLS pooling) $\rightarrow$ MiniLMv2-L6-H768 (66M, $d_S = 768$, 6 layers), a different teacher family; (c) Qwen3-Embedding-0.6B ($d_T = 1024$, last-token pooling) $\rightarrow$ MiniLMv2-L6-H384 (22M, $d_S = 384$, 6 layers), the compact-student setting. In every pair $d_T > d_S$, so the fixed map $P_T = P_{\text{PCA}} R^*$ of Eq. 8 is required.

Training corpus. All methods are distilled on the same corpus of **102,361 unlabeled sentences**, sampled once from the training splits of the nine evaluation benchmarks (SciTail 23,795; Tweet 23,726; Emotion 15,956; STS-B 9,196; Banking77 9,003; MRPC 7,052; WiC 6,333; SICK 4,802; STS12 2,498) and fixed across every run in Table 1. Only raw text is read: no label, pair annotation, or similarity score enters any objective, so the training signal is unsupervised apart from the teacher. The teacher is run once over the corpus before training and its $102,361 \times d_T$ normalized output embeddings are cached; no teacher hidden state, attention map, or forward pass is used during optimization.

Optimization. Students are trained for 5 epochs with AdamW under mixed precision, a cosine schedule with 6% warmup decaying to 2×10^{-6} , a maximum sequence length of 256 tokens, and seed 42. Pooling is CLS at every depth, which matches the pooling used by the evaluation code, so the supervised endpoint is exactly the embedding that is scored. Per-pair learning rates and batch sizes are given in Appendix C.

Reported configuration. The rows labeled GeoODE-KD in Table 1 use the endpoint and contrastive terms of Eq. 29 with $\lambda_{\text{end}} = 1$, $\lambda_{\text{ctr}} = 0.5$, and $\lambda_{\text{evel}} = 0$; the contrastive temperature is $\tau_c = 0.05$ and the second view is an independent dropout mask. The target map is fitted once before training — PCA to d_S on the cache, then the Procrustes rotation R^* of Eq. 7 estimated on 16,384 corpus sentences against the untrained student — and both factors are frozen for the whole run. λ_{ctr} is therefore the only tuned weight, and no parameter is added to the student at any point.

Baselines. We compare against Stella-style multi-dimension output distillation (Zhang et al., 2024), EMO’s optimal-transport embedding objective (Truong et al., 2025), TALAS multi-layer teacher anchoring (Dao et al., 2026), and two cross-tokenizer transfer methods, CDM (contextual dynamic mapping) and DSKD (dual-space knowledge distillation). Each baseline follows its own published recipe, including TALAS’s SAM optimizer, which we leave in place rather than substituting AdamW. The *Teacher* row scores the teacher embeddings directly and the *Student base* row scores the student checkpoint before distillation; both bound the range a distillation method can move within.

Evaluation. We score frozen student embeddings on nine benchmarks in three families. For classification (Banking77, Emotion, Tweet) a logistic-regression probe is fitted on that benchmark’s own training split and scored with macro-F1. For pair classification (MRPC, SciTail, WiC) the cosine similarity is mapped to $[0, 1]$ and scored with average precision. For STS (SICK, STS12, STS-B) we report Spearman correlation against gold scores. The Avg. column is the unweighted mean of the nine primary scores. Appendix D gives the split sizes and the threshold protocol.

Table 1: Main downstream results across three teacher–student configurations. Classification datasets are evaluated with F1, pair-classification datasets with average precision (AP), and semantic textual similarity datasets with Spearman correlation. Higher is better. Within each configuration, the best result among distillation methods is shown in bold and the second-best result is underlined; Teacher and Student base are excluded from this ranking. All distilled students are trained on the same 102,361 unlabeled sentences; the GeoODE-KD rows use the endpoint and contrastive terms ($\lambda_{\text{end}} = 1$, $\lambda_{\text{ctr}} = 0.5$, $\lambda_{\text{vel}} = 0$). Avg. is the unweighted mean of the nine benchmark scores.

Method	Classification (F1)			Pair Classification (AP)			STS (Spearman)			Avg.
	Banking77	Tweet	Emotion	MRPC	SciTail	WiC	SICK	STS12	STS-B	
<i>(a) Qwen3-Embedding-4B $\rightarrow$ BERT-base (109M)</i>										
Teacher	93.63	72.22	68.94	84.79	91.77	66.80	83.43	82.55	86.87	81.22
Student base	84.86	47.79	68.02	77.36	67.74	59.22	42.43	21.54	20.30	54.36
Stella	91.08	73.40	64.40	<u>86.23</u>	83.22	<u>68.98</u>	78.38	71.31	79.16	77.35
CDM	89.71	70.76	52.67	84.03	71.47	69.83	71.36	68.87	74.68	72.60
DSKD	88.89	69.31	53.39	86.35	67.73	68.89	70.58	67.99	74.29	71.94
EMO	85.92	69.39	55.41	85.79	64.34	67.73	63.79	64.41	70.14	69.60
TALAS	93.34	<u>74.45</u>	72.70	86.16	87.78	68.11	<u>80.70</u>	77.34	83.57	80.46
GeoODE-KD	<u>93.22</u>	74.53	<u>72.39</u>	85.92	<u>86.13</u>	67.01	81.17	<u>75.58</u>	<u>81.27</u>	<u>79.69</u>
<i>(b) BGE-M3 $\rightarrow$ MiniLMv2-H768 (66M)</i>										
Teacher	93.52	73.85	68.56	85.81	91.87	61.47	79.18	78.73	84.87	79.76
Student base	81.84	47.70	71.67	79.67	65.06	60.89	49.46	26.76	24.12	56.35
Stella	86.62	71.08	58.57	83.63	75.85	69.38	70.87	63.08	67.88	71.89
CDM	87.96	71.31	58.31	83.83	75.43	69.58	67.37	62.12	67.09	71.44
DSKD	88.45	70.93	58.96	83.25	75.96	69.53	66.19	61.50	65.53	71.14
EMO	87.61	71.19	59.27	82.90	75.03	69.69	65.93	57.46	62.75	70.20
TALAS	<u>90.14</u>	<u>74.91</u>	<u>63.50</u>	<u>86.76</u>	<u>81.27</u>	66.53	<u>76.31</u>	<u>67.66</u>	<u>78.18</u>	<u>76.14</u>
GeoODE-KD	92.37	76.70	66.93	86.96	86.40	63.53	78.84	72.56	80.32	78.29
<i>(c) Qwen3-Embedding-0.6B $\rightarrow$ MiniLMv2-H384 (22M)</i>										
Teacher	93.21	71.33	66.53	83.37	90.00	66.85	80.64	77.11	84.60	79.29
Student base	68.29	41.07	69.90	78.39	64.56	59.58	47.60	21.95	22.08	52.60
Stella	84.36	68.81	54.01	81.61	72.69	<u>68.53</u>	65.30	61.02	64.70	69.00
CDM	85.21	69.36	52.31	81.65	73.12	<u>68.52</u>	65.20	61.91	65.23	69.17
DSKD	85.28	69.34	53.41	81.40	72.77	68.56	64.88	61.42	64.85	69.10
EMO	84.40	69.20	56.13	81.68	72.21	68.22	63.82	57.88	63.34	68.54
TALAS	<u>86.67</u>	<u>72.41</u>	<u>59.93</u>	<u>84.75</u>	<u>80.47</u>	66.72	<u>71.59</u>	<u>69.80</u>	<u>74.94</u>	<u>74.14</u>
GeoODE-KD	91.44	74.24	66.59	85.28	82.47	67.03	77.19	72.33	77.71	77.14

4.2 MAIN RESULTS

Table 1 reports the three configurations. On both MiniLM students the method is the strongest of the six on the average and on eight of nine individual benchmarks (all but WiC): it reaches **78.29** against 76.14 for TALAS on pair (b) (+2.15) and **77.14** against 74.14 on pair (c) (+3.00), while the best non-anchoring baseline stays near 71.9 and 69.2 respectively. On pair (a), the capacity-gap setting, it reaches 79.69 and places second behind TALAS (80.46, -0.77) while remaining 2.34 points above the strongest remaining baseline. Relative to the teacher, the distilled students recover 98.1%, 98.2%, and 97.3% of the teacher average on (a), (b), and (c), starting from student checkpoints at 54.36, 56.35, and 52.60.

The per-task pattern is consistent across pairs. The gains concentrate on STS and on the classification probes — on pair (c), +5.60 SICK, +2.53 STS12, and +2.77 STS-B over TALAS, with +4.77 Banking77 and +6.66 Emotion — and the one family where the method does not lead is WiC, where the projection-based baselines score 68.5–69.8 against 63.5–67.0 here. The same trade-off appears in pair (a), where the residual gap to TALAS comes almost entirely from STS12 (-1.76) and STS-B (-2.30); on SICK and Tweet the ordering reverses.

Overlap between the corpus and two test splits. Because the corpus is sampled from benchmark training splits, some test sentences recur in it as raw text: measured by exact match, 74.5% of SICK test sentences and 21.2% of STS-B test sentences appear in the corpus, while the remaining seven splits are at or below 9% and STS12 is clean. No labels are involved, but the two affected

scores partly measure teacher agreement on memorized text. Dropping SICK and STS-B leaves the conclusions unchanged: the seven-task average on pair (c) is 77.05 for our configuration against 74.39 for TALAS (+2.66), on pair (b) 77.92 against 75.82 (+2.10), and on pair (a) 79.25 against 79.98 (−0.73).

5 DISCUSSION AND LIMITATIONS

6 CONCLUSION

AI USE STATEMENT

ETHICS STATEMENT

REPRODUCIBILITY STATEMENT

REFERENCES

- Ricky T. Q. Chen and Yaron Lipman. Flow matching on general geometries. In *International Conference on Learning Representations*, 2024.
- Ricky T. Q. Chen, Yulia Rubanova, Jesse Bettencourt, and David Duvenaud. Neural ordinary differential equations. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- Quoc Phong Dao, Hoang Son Nguyen, Khanh Chi Pham, Linh Ngo Van, Diep Thi-Ngoc Nguyen, Thien Huu Nguyen, and Trung Le. TALAS: Teacher-anchored layer alignment with adaptive sharpness-aware minimization for embedding distillation, 2026. Manuscript.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Tao Huang, Yuan Zhang, Mingkai Zheng, Shan You, Fei Wang, Chen Qian, and Chang Xu. Knowledge diffusion for distillation. In *Advances in Neural Information Processing Systems*, 2023.
- Bei Li, Quan Du, Tao Zhou, Yi Jing, Shuhan Zhou, Xin Zeng, Tong Xiao, Jingbo Zhu, Xuebo Liu, and Min Zhang. ODE transformer: An ordinary differential equation-inspired model for sequence generation. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*, pp. 8335–8351, 2022.
- Xianming Li, Zongxi Li, Jing Li, Haoran Xie, and Qing Li. ESE: Espresso sentence embeddings. In *International Conference on Learning Representations*, 2025.
- Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matthew Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023.
- Nils Reimers and Iryna Gurevych. Making monolingual sentence embeddings multilingual using knowledge distillation. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing*, pp. 4512–4525, 2020.
- Peter H. Schönemann. A generalized solution of the orthogonal Procrustes problem. *Psychometrika*, 31(1):1–10, 1966.
- Shitong Shao, Zhiqiang Shen, Linrui Gong, Huanran Chen, and Xu Dai. Precise knowledge transfer via flow matching. *arXiv preprint arXiv:2402.02012*, 2024.
- Minh-Phuc Truong, Hai An Vu, Tu Vu, Diep Thi-Ngoc Nguyen, Linh Ngo Van, Thien Huu Nguyen, and Trung Le. EMO: Embedding model distillation via intra-model relation and optimal transport alignments. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, 2025.
- Jiahao Xu, Wei Shao, Lihui Chen, and Lemao Liu. DistillCSE: Distilled contrastive learning for sentence embeddings. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 8153–8165, 2023.

Dun Zhang, Jiacheng Li, Ziyang Zeng, and Fulong Wang. Jasper and stella: Distillation of sota embedding models. *arXiv preprint arXiv:2412.19048*, 2024.

A MATHEMATICAL DERIVATIONS AND PROOFS

A.1 RIEMANNIAN GEOMETRY AND VECTOR FIELD DERIVATIONS

The batch state lies on the product manifold

$$\mathcal{M} = (\mathbb{S}^{d_S-1})^B, \quad (31)$$

equipped with the Frobenius inner product. Its tangent space at $\mathbf{Z}$ is

$$T_{\mathbf{Z}}\mathcal{M} = \{\mathbf{U} \in \mathbb{R}^{B \times d_S} : \mathbf{z}_i^\top \mathbf{u}_i = 0 \text{ for every } i\}. \quad (32)$$

The row-wise projection in Eq. 15 follows directly:

$$\Pi_{\mathbf{Z}}(\mathbf{U})_i = \mathbf{u}_i - (\mathbf{z}_i^\top \mathbf{u}_i) \mathbf{z}_i. \quad (33)$$

Point-wise semantic gradient. For one normalized pair $(\mathbf{z}, \boldsymbol{\tau})$, let $c = \mathbf{z}^\top \boldsymbol{\tau}$ and $\theta = \arccos(c) \in [0, \pi)$. We exclude the antipodal cut locus $\theta = \pi$, where the shortest geodesic is not unique. The ambient derivative of the squared geodesic potential is

$$\nabla_{\mathbf{z}} \frac{1}{2} \theta^2 = -\frac{\theta}{\sin \theta} \boldsymbol{\tau}. \quad (34)$$

Projecting this derivative onto the tangent space gives

$$\begin{aligned} \text{grad}_{\mathbb{S}} \frac{1}{2} \theta^2 &= -\frac{\theta}{\sin \theta} (\boldsymbol{\tau} - c\mathbf{z}) \\ &= -\text{Log}_{\mathbf{z}}(\boldsymbol{\tau}). \end{aligned} \quad (35)$$

The continuous extension at $\theta = 0$ is zero. Applying Eq. 35 to the batch mean in Eq. 12 yields

$$\text{grad}_{\mathbb{S}} \mathcal{E}_{\text{sem}}(\mathbf{Z}, \mathbf{T}) = -\frac{1}{B} \text{Log}_{\mathbf{Z}}(\mathbf{T}). \quad (36)$$

The norm of each logarithmic-map row equals its geodesic distance because

$$\|\boldsymbol{\tau} - c\mathbf{z}\|_2 = \sin \theta, \quad \|\text{Log}_{\mathbf{z}}(\boldsymbol{\tau})\|_2 = \theta. \quad (37)$$

Relational gradient. Let

$$\mathbf{A} = \mathbf{Z}\mathbf{Z}^\top - \mathbf{G}_T. \quad (38)$$

Both Gram matrices are symmetric, so $\mathbf{A} = \mathbf{A}^\top$. For a perturbation $\mathbf{H}$, the differential of Eq. 13 is

$$\begin{aligned} D\mathcal{E}_{\text{geo}}[\mathbf{H}] &= \frac{2}{B^2} \langle \mathbf{A}, \mathbf{H}\mathbf{Z}^\top + \mathbf{Z}\mathbf{H}^\top \rangle_F \\ &= \frac{4}{B^2} \langle \mathbf{A}\mathbf{Z}, \mathbf{H} \rangle_F. \end{aligned} \quad (39)$$

Hence the ambient and Riemannian gradients are

$$\begin{aligned} \nabla_{\mathbf{Z}} \mathcal{E}_{\text{geo}} &= \frac{4}{B^2} \mathbf{A}\mathbf{Z}, \\ \text{grad}_{\mathbb{S}} \mathcal{E}_{\text{geo}} &= \frac{4}{B^2} \Pi_{\mathbf{Z}}(\mathbf{A}\mathbf{Z}). \end{aligned} \quad (40)$$

Combining Eqs. 36 and 40 gives

$$\text{grad}_{\mathbb{S}} \mathcal{E} = -\frac{\alpha}{B} \text{Log}_{\mathbf{Z}}(\mathbf{T}) + \frac{4\beta}{B^2} \Pi_{\mathbf{Z}}[(\mathbf{Z}\mathbf{Z}^\top - \mathbf{G}_T)\mathbf{Z}]. \quad (41)$$

Multiplying by $-B$ recovers the vector direction in Eq. 17. This scaling removes the $1/B$ from the batch mean without changing the descent direction.

A.2 FINITE-HORIZON FLOW DERIVATION

Consider first the teacher-conditioned gradient flow in an auxiliary time τ ,

$$\frac{d\mathbf{Z}}{d\tau} = \mathbf{V}(\mathbf{Z}, \mathbf{T}, \mathbf{G}_T) = -B \text{grad}_{\mathbb{S}} \mathcal{E}. \quad (42)$$

This flow generally reaches a stationary point only as $\tau \rightarrow \infty$. To fit that horizon into student depth $t \in [0, 1]$, define

$$\tau(t) = \int_0^t \frac{s(u)}{R(u)} du, \quad R(t) = \int_t^1 s(u) du. \quad (43)$$

Since $\dot{R}(t) = -s(t)$,

$$\frac{d}{dt} \log R(t) = -\frac{s(t)}{R(t)}, \quad (44)$$

and therefore

$$\tau(t) = \log \frac{R(0)}{R(t)}. \quad (45)$$

For any schedule with $R(t) > 0$ for $t < 1$ and $R(1) = 0$, $\tau(t) \rightarrow \infty$ as $t \rightarrow 1$. Applying the chain rule to Eq. 42 gives

$$\frac{d\mathbf{Z}}{dt} = \frac{d\tau}{dt} \frac{d\mathbf{Z}}{d\tau} = \frac{s(t)}{R(t)} \mathbf{V}(\mathbf{Z}, \mathbf{T}, \mathbf{G}_T), \quad (46)$$

which is Eq. 19.

Schedules. For $s(t) = t^p$ with $p > -1$,

$$R(t) = \frac{1 - t^{p+1}}{p+1}. \quad (47)$$

The constant schedule is $p = 0$, while the default linear schedule is $p = 1$ and gives $R(t) = (1 - t^2)/2$.

Discrete step fraction. The warped time elapsed between two student depths is

$$\begin{aligned} \Delta\tau_\ell &= \int_{t_\ell}^{t_{\ell+1}} \frac{s(u)}{R(u)} du \\ &= \log \frac{R(t_\ell)}{R(t_{\ell+1})}. \end{aligned} \quad (48)$$

Under the instance-only auxiliary flow, the distance remaining after this interval is multiplied by $\exp(-\Delta\tau_\ell)$. The fraction covered is therefore

$$\begin{aligned} \rho_\ell &= 1 - \exp(-\Delta\tau_\ell) \\ &= 1 - \frac{R(t_{\ell+1})}{R(t_\ell)}, \end{aligned} \quad (49)$$

which recovers Eq. 21. For the default schedule and $t_\ell = \ell/L$,

$$\rho_\ell = \frac{2\ell + 1}{L^2 - \ell^2}, \quad \rho_{L-1} = 1. \quad (50)$$

For the point-wise field, this fraction reproduces the exact geodesic progress over the interval. For the coupled semantic-relational field, GeoODE-KD uses the same ρ_ℓ as a local exponential-map step; it is a discretization, not an exact integration of the changing nonlinear field.

The exponential map also preserves the discrete target norm. If $\mathbf{v} \in T_{\mathbf{z}}\mathbb{S}^{d_S-1}$, then $\mathbf{z}^\top \mathbf{v} = 0$, and Eq. 23 gives

$$\|\text{Exp}_{\mathbf{z}}(\mathbf{v})\|_2^2 = \cos^2 \|\mathbf{v}\|_2 + \sin^2 \|\mathbf{v}\|_2 = 1. \quad (51)$$

A.3 PROOFS OF THE FLOW PROPERTIES

The following statements assume $s(t) \geq 0$, $R(t) > 0$ for $t < 1$, and that no student–teacher pair lies at the antipodal cut locus. The logarithmic map is defined by continuous extension when a student state already equals its target.

Proposition 1 (norm preservation). Every row of a solution to Eq. 19 remains on the unit hypersphere.

Proof. Both terms in Eq. 17 are tangent: the log map lies in $T_{\mathbf{z}_i} \mathbb{S}^{d_S-1}$, and the relational term is explicitly projected by $\Pi_{\mathbf{z}}$. Thus $\mathbf{z}_i^\top \dot{\mathbf{z}}_i = 0$ and

$$\frac{d}{dt} \|\mathbf{z}_i(t)\|_2^2 = 2\mathbf{z}_i(t)^\top \dot{\mathbf{z}}_i(t) = 0. \quad (52)$$

Since $\|\mathbf{z}_i(0)\|_2 = 1$, its norm remains one. $\square$

Proposition 2 (energy descent). Along the ideal continuous flow, the teacher-conditioned potential is non-increasing for every $t < 1$.

Proof. By Eq. 41, $\mathbf{V} = -B \text{grad}_{\mathbb{S}} \mathcal{E}$. The Riemannian chain rule gives

$$\begin{aligned} \frac{d\mathcal{E}}{dt} &= \left\langle \text{grad}_{\mathbb{S}} \mathcal{E}, \dot{\mathbf{Z}} \right\rangle_F \\ &= -\frac{B s(t)}{R(t)} \|\text{grad}_{\mathbb{S}} \mathcal{E}\|_F^2 \leq 0. \end{aligned} \quad (53)$$

This proves Eq. 30. $\square$

Corollary 1 (point-wise geodesic contraction). For $\alpha = 1$ and $\beta = 0$, the geodesic distance $\theta(t) = d_g(\mathbf{z}(t), \boldsymbol{\tau})$ satisfies

$$\theta(t) = \theta(0) \frac{R(t)}{R(0)}, \quad \lim_{t \rightarrow 1} \theta(t) = 0. \quad (54)$$

Proof. For one pair, define $e(t) = \frac{1}{2}\theta(t)^2$. From Eq. 35 and the point-wise form of Eq. 19,

$$\begin{aligned} \dot{e}(t) &= \left\langle -\text{Log}_{\mathbf{z}}(\boldsymbol{\tau}), \frac{s(t)}{R(t)} \text{Log}_{\mathbf{z}}(\boldsymbol{\tau}) \right\rangle \\ &= -\frac{s(t)}{R(t)} \theta(t)^2, \end{aligned} \quad (55)$$

where Eq. 37 is used in the last line. For $\theta(t) > 0$, $\dot{e} = \theta \dot{\theta}$, so

$$\frac{d}{dt} \log \theta(t) = -\frac{s(t)}{R(t)} = \frac{d}{dt} \log R(t). \quad (56)$$

Integrating from 0 to t proves Eq. 54. If θ reaches zero earlier, the log-map vector is zero and the state remains at the target. $\square$

If the observed student trajectory begins at the first Transformer layer $t_1 = 1/L$ rather than at $t = 0$, the same argument starts from t_1 and gives

$$\theta(t) = \theta(t_1) \frac{R(t)}{R(t_1)}, \quad t \geq t_1. \quad (57)$$

Since $\theta(t) \in [0, \pi)$ and $\dot{\theta}(t) \leq 0$, teacher cosine is also non-decreasing:

$$\frac{d}{dt} \cos \theta(t) = \frac{s(t)}{R(t)} \theta(t) \sin \theta(t) \geq 0. \quad (58)$$

Corollary 2 (exact sampled profile). For $0 \leq a \leq 1$, the curve

$$\gamma(a) = \text{Exp}_{\mathbf{z}}(a \text{Log}_{\mathbf{z}}(\boldsymbol{\tau})) \quad (59)$$

is the shortest geodesic from $\mathbf{z}$ to $\boldsymbol{\tau}$ and satisfies

$$d_g(\gamma(a), \boldsymbol{\tau}) = (1 - a)d_g(\mathbf{z}, \boldsymbol{\tau}). \quad (60)$$

Setting $a = \rho_\ell$ in the instance-only target gives

$$\theta_{\ell+1} = (1 - \rho_\ell)\theta_\ell = \frac{R(t_{\ell+1})}{R(t_\ell)}\theta_\ell. \quad (61)$$

The factors telescope, so the predicted states reproduce Eq. 54 at every sampled depth and the last target equals $\boldsymbol{\tau}$ because $\rho_{L-1} = 1$.

Scope of the guarantees. Norm preservation and energy descent describe the ideal continuous field. Exact arrival and the sampled profile require the instance-only setting $\alpha = 1, \beta = 0$. With $\beta > 0$, the native teacher Gram target can compete with the projected point-wise target, so individual distances need not decrease at every depth. Finally, the realized Transformer trajectory satisfies the sampled profile only to the extent that it minimizes $\mathcal{L}_{\text{eval}}$; the propositions do not assume zero discretization or optimization error.

B TARGET CONSTRUCTION AND GAUGE ALIGNMENT

B.1 PCA INITIALIZATION

B.2 GRAM-CONSISTENT TARGET FITTING

B.3 ORTHOGONAL GAUGE ALIGNMENT

C TRAINING ALGORITHM AND IMPLEMENTATION DETAILS

C.1 HYPERPARAMETERS

Table 2 lists every setting behind Table 1. All three pairs share one corpus, one schedule shape, and one objective configuration; only the learning rate, batch size, and the widths that follow from the pair itself differ.

C.2 COST

The teacher is run once over the corpus and its normalized outputs are stored as a single $102,361 \times d_T$ float32 tensor (0.4 GB at $d_T = 1024$, 1.0 GB at $d_T = 2560$). Fitting P_{PCA} is one truncated SVD of the cache; fitting R^* is one $d_S \times d_S$ SVD over 16,384 sentences. Both are computed before the first optimizer step and never updated, so training itself sees only a fixed lookup table of targets and never loads the teacher.

C.3 HYPERPARAMETER MATCHING ACROSS METHODS

The baseline rows follow the recipe published with each method rather than a single shared optimizer setting: TALAS keeps its SAM optimizer, and the cross-tokenizer baselines keep the learning rate and batch size of their reference implementations. The comparison in Table 1 is therefore between methods as they are normally run, not between objectives at matched optimization settings, and part of the reported margin may be attributable to that difference.

D DATASETS, BASELINES, AND EVALUATION PROTOCOL

D.1 BENCHMARKS

Table 3 lists the nine evaluation benchmarks, the split sizes, and the primary metric of each family. The same nine benchmarks supply the raw text of the distillation corpus (Section 4.1), sampled from their training splits only.

Table 2: Hyperparameters of the reported runs. The objective block gives the configuration used for every GeoODE-KD row in Table 1.

	(a) Qwen3-4B → BERT-base	(b) BGE-M3 → MiniLM-H768	(c) Qwen3-0.6B → MiniLM-H384
<i>Data</i>			
Training sentences	102,361 (unlabeled)		
Max sequence length	256		
Teacher width d_T	2560	1024	1024
Student width d_S	768	768	384
Student layers L	12	6	6
Teacher pooling	last token	CLS	last token
Student pooling	CLS		
<i>Optimization</i>			
Optimizer	AdamW		
Epochs	5		
Schedule	cosine, 6% warmup, floor 2×10^{-6}		
Precision	mixed (fp32 weights, autocast)		
Seed	42		
Learning rate	TODO	TODO	5×10^{-5}
Batch size	TODO	TODO	64
<i>Objective</i>			
λ_{end}	1.0		
λ_{evel}	0.0		
λ_{ctr}	0.5		
Contrastive temperature τ_c	0.05		
Second view	independent dropout mask		
<i>Target map $P_T = P_{\text{PCA}} R^*$</i>			
PCA fit	on the cached corpus embeddings, centered		
Gauge alignment	Procrustes on 16,384 sentences, fitted once		
Refit frequency	never (frozen after fitting)		

Table 3: Evaluation benchmarks. The probe column gives the number of examples used to fit the logistic-regression classifier; pair and STS benchmarks need no fitting. Corpus is the number of sentences the benchmark contributes to the 102,361-sentence distillation corpus.

Family	Benchmark	Probe train	Test	Corpus	Primary metric
Classification	Banking77 (77 classes)	9,016	3,076	9,003	macro-F1
	Emotion (6 classes)	15,956	1,986	15,956	macro-F1
	Tweet (3 classes)	23,726	3,432	23,726	macro-F1
Pair classification	MRPC	—	1,725	7,052	average precision
	SciTail	—	2,126	23,795	average precision
	WiC	—	1,400	6,333	average precision
STS	SICK	—	4,927	4,802	Spearman
	STS12	—	3,108	2,498	Spearman
	STS-B	—	1,379	9,196	Spearman

D.2 PROTOCOL

Every method is scored the same way, on frozen student embeddings with CLS pooling and no fine-tuning of the encoder. Classification benchmarks fit a logistic-regression probe on the benchmark’s own training split and score the test split; the probe train and validation files are checked for normalized-text leakage against the test split before fitting. Pair benchmarks map the cosine similarity of the two sentences to $[0, 1]$; we report average precision, which ranks pairs and therefore needs no decision threshold. STS benchmarks correlate the raw cosine similarity with the gold score (Spearman). The Avg. column of Table 1 is the unweighted mean of the nine primary scores.

756 The reported runs evaluate the test split after every epoch with no separate validation pass, and the
757 reported numbers are those of the final (fifth) epoch; no checkpoint is selected on the basis of a test
758 score. Threshold-dependent accuracy and F1 for the pair family are produced by the same harness
759 with the threshold swept on the evaluation split, and are therefore not reported here.
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761 E ADDITIONAL RESULTS AND DEPTH DIAGNOSTICS
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763 F EXTENDED ABLATIONS AND SENSITIVITY ANALYSIS
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